

Outer Race Bearing Replacement

Electric Motor — Stage 3 Outer Race Fault
Assets: AST_MTR_001 (Conveyor Motor A) | Asset also covers AST_MTR_002 frame
Bearing Type: SKF6310 | Est. Duration: 6 h | Crew: 2 | Priority: HIGH/URGENT
SOP_001 | v1.2 | Revised: 2026-03-10

1. Purpose and Scope

This procedure covers planned replacement of a drive-end or non-drive-end deep groove ball bearing on medium-to-large industrial electric motors where an outer race fault has been confirmed at ISO 13373-1 Stage 3 or above. Stage 3 is defined as BPFO band energy exceeding 3.5× the established healthy baseline AND kurtosis exceeding 5.0. At Stage 3, bearing replacement is mandatory — lubrication or re-greasing will NOT arrest degradation. Applies to SKF6310-series bearings on Siemens SIM-MTR-400 (AST_MTR_001) and ABB ABB-MTR-220 (AST_MTR_002) frames in the 45–75 kW range.

2. Prerequisites

- Maintenance window approved and active in scheduling system (e.g. CTX_001 on LINE_001, 4 h minimum)
- Work order raised, approved, priority confirmed HIGH or URGENT
- PART_001 (SKF6310-ZZ bearing) confirmed available in BIN_A1, quantity at least 1
- PART_004 (Synthetic Grease) confirmed available in BIN_L1
- LOTO permit LOTO_001 confirmed current (is_current = true, expiry date not passed)
- For AST_MTR_002: LOTO_005 must be confirmed current
- Two technicians available, both LOTO_CERTIFIED
- At least one technician holding ISO_CAT_III or CBM_LEVEL_2 for post-repair validation
- Bearing puller, torque wrench, induction heater, and portable vibration meter on site

3. Safety Requirements

3.1 LOTO Isolation Procedure

LOTO Permit	LOTO_001 (AST_MTR_001) LOTO_005 (AST_MTR_002)
Isolation Points	MCC-12, Valve-4 (AST_MTR_001) / MCC-13, Valve-5 (AST_MTR_002)
Energy Types	Electrical
Isolation Steps	Open and lock MCC. Apply personal danger tag. Open valve and apply lock. Each technician applies their own padlock.
Verification	After isolation, attempt to start motor at control panel. Confirm motor does not start. Confirm zero voltage at terminals using approved voltage tester.

3.2 PPE Requirements

Mandatory PPE	Electrical insulation gloves (Class 00 minimum), safety helmet, steel-toed boots, vibration-damping gloves, safety glasses
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3.3 ISO 10816-3 Zone Check

Zone — AST_MTR_001	C — Controlled approach permitted with standard PPE. Do not operate at elevated load until repair is complete.
Zone — AST_MTR_002	A — Normal operating range; standard approach applies.

4. Tools and Materials Required

Part Number	Description	Quantity	Bin Location
PART_001	SKF6310-ZZ Deep Groove Ball Bearing	1	BIN_A1
PART_004	Synthetic Grease (Shell / equivalent)	1 unit	BIN_L1

- SKF bearing puller TMBS 30E or equivalent mechanical puller
- Torque wrench, calibrated, range 20–150 Nm
- Induction bearing heater (target 110°C for mounting)
- Infrared thermometer
- Portable vibration analyser — mm/s RMS capability
- Clean lint-free rags and bearing cleaning solvent
- Feeler gauge set for clearance check
- Bearing installation sleeve or press plates matching SKF6310 bore

5. Step-by-Step Procedure

1	Confirm work order approved, maintenance window active, and parts on site before touching the machine.
2	Isolate the motor: apply LOTO per Section 3.1. Each technician fits personal lock.
3	Verify zero energy: attempt start at control panel. Confirm zero voltage at motor terminals.
4	Remove motor end cover (drive end or non-drive end as applicable). Document cover bolt torques found.
5	Using bearing puller TMBS 30E, extract the bearing from the shaft journal. Note resistance and any abnormal sounds during removal.
6	Inspect removed bearing: photograph spalling pattern, measure raceway damage extent, note grease condition (colour, smell, particle content). Record findings on work order.
7	Inspect shaft journal OD and housing bore ID with feeler gauges. Confirm both within drawing tolerance. If out of tolerance, STOP and escalate.
8	Inspect bearing housing end cover seal. If cracked or degraded, replace seal — failure to replace allows re-contamination.
9	Clean housing bore and shaft journal with lint-free cloth and approved solvent. Remove all old grease and metallic debris.
10	Heat new SKF6310-ZZ bearing in induction heater to 110°C. Do not exceed 125°C.
11	Slide heated bearing onto shaft journal smoothly — no hammering. Apply gentle press force only to inner ring. Allow to cool to ambient.
12	Apply PART_004 Synthetic Grease to approximately 30–40% of bearing cavity volume. Do not over-grease.
13	Refit end cover. Torque cover bolts to 45 Nm in cross pattern.
14	Remove LOTO: each technician removes their own lock. Confirm all tools and rags removed from machine.
15	Controlled restart: bring machine to operating speed at reduced load (50%) for first 30 minutes.
16	Measure vibration RMS with portable meter at 30 minutes. Record reading on work order.
17	Return to full operating load. Take second vibration and temperature reading at 1 hour.
18	QA check: confirm post-repair readings meet criteria in Section 7. Record all measurements.
19	Final QA check at 24 hours post-restart: vibration, temperature, and kurtosis from historian.
20	Close work order. Update bearing_master.last_lubrication_date and current_health_score.

6. Critical Specifications

Torque Specification	End cover bolts: 45 Nm in cross pattern
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Bearing Fit	Interference fit on shaft journal — induction heating method required
Mounting Temperature	110°C target, max 125°C
Lubricant Type	PART_004 — Synthetic Grease (Shell Gadus S2 V220 or equivalent)
Lubricant Quantity	30–40% of bearing cavity volume (approx. 45–60 g for SKF6310 frame)
Fill Level	Do not exceed 50% — over-greasing causes churning and thermal rise

7. Post-Repair Quality Acceptance Criteria

All values must be met at BOTH the 1-hour and 24-hour post-restart checks. If any criterion fails at 1 hour, do not proceed to full load — investigate immediately.

Parameter	Acceptance Criterion	Failure Action
Vibration RMS (at rated speed, full load)	$\leq 2.5 \text{ mm/s}$	Shut down, re-inspect bearing seating
Peak Vibration	$\leq 3.0 \text{ g}$	Shut down, re-inspect
Bearing Temperature	$\leq 60^\circ\text{C}$ at 1 h $\leq 55^\circ\text{C}$ at 24 h	Check lubricant fill level; re-inspect if temperature rising
Kurtosis (at 24 h)	≤ 2.8	Re-inspect bearing seating and shaft alignment
BPFO Energy (normalised)	$\leq 1.2 \times \text{baseline}$	Bearing seating suspect; escalate to maintenance lead
Signal Quality Score	≥ 0.97	Check sensor cable and channel CH_001 / CH_002 connection

8. Common Failure Modes and Troubleshooting

- If vibration does not fall below 2.5 mm/s after restart → bearing may be misseated; shut down, remove and re-press.
- If temperature rises above 70°C within 1 hour → excess grease causing churning; drain and re-pack to 35% fill.
- If kurtosis remains above 3.0 at 24 h → check bearing inner race for fit on shaft journal; may require shaft repair.
- If BPFO energy remains above 1.5× at 24 h → confirm correct bearing model was installed; wrong model produces false signals.
- If housing end cover seal was not replaced and fault recurs within 90 days → contamination re-entry; replace seal and re-inspect.

9. Related Documents

- SOP_006 — Post-Repair QA Procedure (all assets)
- LOTO_001 permit record — AST_MTR_001 (MCC-12, Valve-4)
- LOTO_005 permit record — AST_MTR_002 (MCC-13, Valve-5)
- SKF6310-ZZ installation guide (OEM bearing installation manual)
- CASE_001 learned case — Stage 3 outer race, AST_MTR_001, contamination confirmed